

Troll Farm — The D-Series Atlas

A reader's guide to every numbered experiment of the Legend top-3 cycle

Snapshot 2026-07-27 · ledger vol 1+2 · docs/CONSTRAINTS.md · docs/STATE.md

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Snapshot date: 2026-07-27. A reader's guide to every numbered experiment of the Legend top-3 research cycle. Chronology lives in the ledger volumes (data/analysis/live-agent-6553250/legend-top3-expo

topic-sorted conclusions in docs/CONSTRAINTS.md; live state in docs/STATE.md. This atlas is a navigation snapshot, not a living record.

1 What a D-experiment is

Each Dnnn is one frozen-protocol experiment in the project's laboratory notebook. A letter suffix (D133b, D41c) is a protocol revision or repair of the same experiment. Every experiment leaves the same artifacts in data/analysis/live-agent-6553250/: a **protocol** (question, method, and pass/fail thresholds written before running), a **lock** (checksums of protocol, code, and inputs), the run products (bulk rows on the external artifacts/ root), and a **result** document ending in an explicit verdict. The discipline that holds it together: thresholds are never adjusted after seeing data, consumed seed ranges are never reused for selection, and negative results are retained as binding constraints. Numbering makes conclusions citable ("remembered-crop returns are closed — D165") and prevents re-running dead ends.

The goal (set 2026-07-18): rank 3 or better in the Legend practice ladder, on a mature read plus confirmation. As of 2026-07-27 the resident (agent 6561795, exact slim Yamo/Orchard, two workers) reads 43/110 at 21.97 against a rank-3 bar of 28.22. The measured disease: the resident leads at turn 100 in most eventual losses and is out-compounded late by opponents who scale to three-plus workers while keeping production alive; top agents reap 24.16% of the crops they create, the resident 0.94% (D101).

2 Prelude — Phases 1-21 (2026-07-16 → 07-18)

Before the D-series, a phase-numbered program recovered the live bot and made the first candidate attempts. Highlights:

Phase	Question	Verdict
Recovery	What is actually live?	Exact source recovered (rank 6/104 @ 26.31), checksummed, made submit default
1-5	Do micro-edits help?	Idle-harvest KEEP; focus bonus KEEP; training constants closed; renewable mothers closed
Terminal fix	Are local verdicts valid?	Referee stall/mercy semantics adopted; one verdict flipped (pre-seed neutral → +0.259)
Stack	Pre-seed + orchard geometry	Arena +3.0 — PROMOTED 2026-07-17; slim A/A passed; still the only arena win
3-5	First-turn rollout controller	Local +2.7 → arena reject (21.7 vs 24.1 control)
6-8	29-option first-move library	No opponent-robust activation; turn-one Monte Carlo closed
9	Imitate Escdemon	Passes observationally; collapses at autoregressive integration (52% MOVE)
10	Norxondor's workforce ladder	Reproduces all 8,738 triggers; observable-signature switch fails prospectively (−6.2)
11	Online shared-state Monte Carlo	Teacher +26.1 but 209 ms vs 50 ms budget — undeployable
12-15	Distill trajectory value	Fails held-opponent-family precision every time; native imitation −172.7

Phase	Question	Verdict
16	Resident MOVE residual	Prospective +0.508 at 93 ms p95 — closed
21	Dual-value opponent-crop scoring	All local gates pass → arena -7.77 REJECT ; b100_e6 evolved candidate +0.12, closed

Lesson carried forward: local gates were not predicting arena outcomes; the cycle moved to architecture research with much harder transfer discipline.

3 Arc A — Curriculum and the deployment bridge (≈D1-D28, 07-19 → 07-20)

Can a neural policy learn this game at all, and can it ship inside CodinGame’s limits? A training curriculum was built in five levels; numbering here is partly internal to the rolling narrative (headered experiments begin at D29).

Item	Question	Verdict
PPO from scratch	Learn the debug task raw	12.4% — closed at readiness
BC + teacher-auxiliary	Bootstrap from a scripted teacher	Pure PPO argmax-collapses (18.2%); a 0.10 teacher anchor rescues it (99.3%)
Level 1	Execute one worker’s resource plan	Accepted, independently replicated (99.65%)
Level 2	All 8 worker recipes	Accepted (98.6%), discovery + confirmation
Level 3	Two-role renewable loop	Accepted (99.45%); illegal-label mask repaired
Level 4	Recipes × roles jointly	Accepted (99.70%)
L5 D1-D5	Isolated opponent mechanisms	Forager, planter, reaper, funded 2-worker: all accepted
L5 D6-D8	Paid third worker, horizons	Teacher-censoring artifacts; closed at controls
L5 D9	Crop before scale	Accepted — “secure renewable supply before scaling”
L5 D10-D11	Repeated pressure; seed reacquisition	D11 = first task to beat the actor (79.4%); clone+PPO accepted at 97.65% prospective
K1-K2	int8 deployment	K2 accepted: 7.04 ms median inference
V1-V5	Live-protocol integration	Four observation-parity defects fixed; V5 accepted (68,988 B, 17.6 ms warm p95)
YT bench-mark	Train remotely?	9.8× throughput but CUDA/CPU parity fail → local training chosen

Surviving assets: the environment/actor infrastructure, the deployment pattern (int8 + persistent buffers), the teacher-auxiliary recipe — and a deployable curriculum actor whose *field* value was still unproven.

4 Arc B — The domain-shift reckoning (D29-D34, 07-20)

ID	Question	Verdict
D29a/b	Turn-75 spatial option-critic (farm vs resident)	+36.5 on generated maps; int8 candidate built
D29c	Does it activate on real arena states?	8.75% vs 17-67% corridor — closed unsubmitted
D30	Why the -78 shift?	Generator artifact: 6 water cells vs official 12-104; all roots out of support
D31	Replay recorded commands as continuations?	Invalid — only 86.85% command match
D32a	TestSession forced-farm A/B	Mean -29; permanent farming and threshold repair closed
D33a/b	Port the referee's real map generator	Accepted — SHA1PRNG + plant-order subtleties; 120/120 exact
D34	Census: 9 complete architectures on official maps	None passes production+suppression; worker count alone $\neq$ strength

This arc invalidated the old evaluation world and replaced it with the authoritative official-map substrate. Everything after D33 runs on real map geometry.

5 Arc C — Job grammars and bounded overlays (D35-D36, 07-20 → 07-21)

ID	Question	Verdict
D35a	Decode rich players' persistent jobs	Vocabulary strong (94% RENEW+FELL); flat categorical head fails
D35b	One-shot factorized two-worker bundles	Oracle +34.99 but suppression far short
D35c	Add crop-creator provenance to targets	Causal +7.8 — provenance retained as a factor; one-shot closed
D35d	Repeat allocation at job boundaries	+23.1 beyond one-shot; opponent excess still +76 vs +65 ceiling
D36	Same overlays anchored on the resident	+19.6 own / +10.6 margin vs +68/+25 floors — closed

Conclusion recorded: no wrapper composition reaches the field gap; one coherent policy must own opening, supply, funding, production, and suppression end to end.

6 Arc D — Complete-macro teachers (D37-D40, 07-21)

ID	Question	Verdict
D37	Complete factorized macro environment	Environment clean and deterministic; rate initializer fails; transactional semantics invalidation found and fixed

ID	Question	Verdict
D38	Value jobs by exact TRAIN-bill reduction	Rejected (loses to random); spawn-shack blocker isolated
D39	Add shack evacuation	Narrow reject (+38.2 vs +50); insight: 1→3 promotions gain +295, 1→2 gain −1.3
D40	Deficit-first, else work-conserving rates	PASSES +94.6 over D39, all families — frozen as THE teacher

7 Arc E — Learning on D40 (D41-D62, 07-21)

The graveyard of straightforward learning ideas, and two validated interfaces.

ID	Question	Verdict
D41a	Behavior-clone D40	85% vs 99% floor; a parameter-free decoder gets 100% — MLPs can't do lexicographic filters
D41b	Exact-prior + zero residual	Byte-exact reproduction — validated interface
D41c	Residual PPO (temp 4)	Max residual +0.33 vs +4 prior gap → 0/85,128 changed decisions
D41d-f	One-deviation labels, branch selectors, thresholds	46-58% positive vs 55-60% floors — all fail
D41g/h	Linear / tiny-ReLU value filters	Fail the 65%/27% precision pair
D42	194-feature context selector	Fails; one-deviation supervision closed
D43/D44	Sparse binary closed-loop PPO	Learns a uniform bias (probe SD 0.0012); its scores uncorrelated with value
D45/D48	Rate-formula surfaces	Saturated/discontinuous — not locally searchable; CEM blocked
D46/D47	Hard chopper / producer roles	Chopper already implicit in D40; producer role −12.0
D49	Chopper-first reservation order	12 deposit-prediction violations — transactional revalidation required
D50	Population of phase-recombined opponents	Support improves; rich-cohort gates still missed
D51	Workforce-relative whole-controller switching	Only 27.5% ever reach the trigger — funding is the upstream problem
D52a/b	Procedural job market; why TRAIN fails	All 5,142 failures = a PICK spends the bill after the decision
D53a/b	Atomic bill reservation	Fixes worker-2; 168 residuals = shared-surplus oversubscription
D54	Shared per-turn PICK ledger	Transactions fully clean; reach still 25.3% — problem moves to acquisition
D55	Terminal stock-flow diagnostic	91% of blocked bills source-unresolved; LEMON dominates
D56/D57	LEMON source; exact deficit-vector sources	Sources get built; worker-3 reach <i>falls</i> (23.8%, 21.3%)
D58	Labor/progress diagnostic	Source planting regresses bills; mining converts — conversion layer is causal

ID	Question	Verdict
D59	Materialization-only lease	Better per-turn progress, 56% idle labor — entire hand-designed source branch closed
D60	Fixed invest/materialize/liquidate per phase	Oracle +47.4 but violates the crop floor — fixed modes rejected
D61	Options at natural job-batch boundaries	Oracle +57.6, crop-safe, sequencing carries value — validated interface
D62	Feed-forward batch-option PPO	Argmax never moves (0/512) — recipe closed; passive field refresh (D61p) readied

8 Arc F — Field evidence, round two (D63-D71, 07-21)

ID	Question	Verdict
D63a/b	What predicts who scales?	Turn-100 economy state (0.97 AUC), not opening geometry (0.48); prediction $\neq$ value
D64a/i	Gate late scale by the field model	Run invalid via a worker-two tail; oracle only +1.9; diagnosis: missing bill species
D65-D67	Rescue via deposited-seed sources	Planted sources die before payback; lease fails; zero viable cells — closed
D68	Bill-level source portfolio	2/4 recovery; reusable hostile pressure taxes each generation — late rescue closed
D69/D70	Scaler lifecycle archaeology	Universal order establish→receipt→reinvest→worker-3; no minimal opening transaction reproduces it
D71	Closed-loop opening-portfolio environment	Mechanics pass — infrastructure only

9 Arc G — Recurrent search era (D72-D78, 07-21)

ID	Question	Verdict
D72	Recurrent opening portfolios (8 actions)	Oracle +64.1 headroom; explicit seed-action breadth misses by one task
D73	Four-mode recurrent PPO	−1.76 prospective; sacrifices own score symmetrically
D74/D75	One- and two-batch option sequences	Too coarse: 38.5% strict; 16 sequences span 3.5 points
D76/D77	CEM; full-parameter lineage search	Both converge to the balanced plateau ($\leq 1\%$ non-balanced choices)
D78a/b	Is opponent attack imminence observable?	Yes — 0.9307 AUC from spatial features; history adds nothing

Recorded pattern: with four coarse modes and a snapshot state, every optimizer retreats to “do what the resident does.” The action space, not the optimizer, was the constraint.

10 Arc H — Spatial and target interventions (D79-D85, 07-21)

ID	Question	Verdict
D79	Unconstrained all-Rate spatial scorer	Global trajectory replacement (every policy changes every hash) — closed
D80/D81	One-shot contested-crop promotions	Ubiquity/support failures; lesson: measure authority by decisions, not hashes
D82	Semantic responses to threatened crops	Oracle +11.2 but value is state-specific — no deployable fixed arm
D83	Snapshot value model over D82	Captures 10.6% of oracle — closed
D84	Truncated counterfactual Monte Carlo	Best +3.2 vs +5.6 needed, at 210 ms — online MC closed
D85	One-turn defensive salvage on real replays	The resident already makes every lethal joint chop; salvage +0.05 — closed

11 Arc I — The yaichi seed-factory (D86-D95, 07-21)

ID	Question	Verdict
D86	Is renewable mode opening-selectable?	No — 0.58 validation accuracy
D87	Commit to regeneration after fresh harvests	−51.2 active margin; new plants become wood, not an orchard
D88a-c	Decode yaichi’s task states	Bank-bootstrap seed-factory architecture confirmed 10/10
D89	Reproduce the factory on the resident	Production spectacular (+79.4) but opponent +82.9 — safety reject
D90	Lineage boundaries; ATTACK state	Ablation inert; ATTACK is an endgame bank blockade
D91	Select factory activation by map	Development +31 but 5/16 maps — transfer fail
D92	Factory + dual-value denial	Trained-role denial arrives too late (−5.6)
D93	Can the factory fund worker 3?	Zero legal TRAIN turns in 256 tasks — it’s a wood economy, not a funding economy
D94	Late existing-stock funding bridge	Trains in 147 tasks and loses 91.6 — late grafts closed
D95	Rank-one scaler archaeology	Coordinated funding is shared; no universal hand-written grammar — learned-representation requirements recorded

12 Arc J — Joint assignments and the q6 program (D96-D147, 07-21 → 07-23)

The largest arc: a validated coordination interface, a compact action bank, and roughly thirty supervised selectors that all failed the same way.

ID	Question	Verdict
D96	Worker-factorized four modes	+0.8 incremental vs +5 — worker context isn't the missing piece
D97	Concrete collision-safe JOINT assignments	Oracle +36.9; joint adds +9.2 beyond best-single — validated interface
D98-D100	Pair scorers, replacements, residuals	Near-misses and activity failures; static selection over the bank is dead (rank correlations 0.10-0.22)
D101	Production/suppression role archaeology	□ The gap: top-3 reap 24.16% of created crops, resident 0.94%; suppression already competitive
D102/D103	Transfer D40 wholesale; phase decomposition	−48.4; opponent excess accrues 14/39/47% across phases — duration matters
D104-D106	Expert-proposal union; q4/q6 quantization	Union keeps 86-88% of the joint oracle; q6 bank frozen at 9,180 bytes with fresh-map replication (+32.0)
D107	Bounded online controller preflight	Four-use oracle +35.2, crops/workforce exact — interface ready
D108/D109	Recurrent masked q6 PPO; 4× duration	−0.74 then −0.15; family patterns rotate across panels — the family-robust objective question is posed (and skipped)
D110/D111	Random linear one-use policies; lineage search	Abstention calibration is the frontier; fitness doesn't transfer
D112/D113	Dense exact-continuation teacher	Coverage semantics fixed; teacher +36.8, all families positive
D114-D118	Ridge, MLP, WAIT-softmax, factorized gates, soft targets	All fail; ranking (not timing) is the bottleneck
D119-D127	Long fits, info floors, calibration, tail attribution	First fit+validation pass appears; held support protocol fails; 82/98 losses are ranking errors with +891 recoverable
D128/D129	Absolute anchors; safety heads	Ranking and safety must not share a scalar; compositions 0/60
D130/D131	Pairwise loss; all-seed audit	Fit statistics <i>anti-predict</i> transfer ($r = +0.89$ wrong way)
D132/D133/D134	Distributed corpora	Byte-exact parity; eight independent 16-map blocks built
D134-D138	Out-of-fit block selection; gates; calibration	Selection design validated; every learner still fails the veto
D140-D143	8-block selection, balanced losses, dual gates	Best ever: +3.06 at 39.65% strict vs 40% floor — one-use gate tuning closed
D144-D146	Offline two-intervention Monte Carlo	+4.15 incremental; winning pairs genuinely joint (second action +27.3); early-immediate schedules carry it
D147	Live-interface feature replay	Byte-exact — safe to scale

13 Arc K — Exact-value teachers and the substrate verdict (D148-D158, 07-22 → 07-23)

ID	Question	Verdict
D148a/b	Fresh-map priority joint teacher (YT)	Hidden two-use transfer +4.11, robust in all families
D149a/b	Imitate its best pairs	9.5–9.8% rank accuracy — hard-argmax entropy, closed
D150	Is near-tie supervision available?	16.5% vs 20% — collect counterfactuals instead
D151/D152	Exact conditional-second values (YT)	+36.8 on active states — the strongest teacher signal of the program
D153–D156	Fitted value, abstention, slices, memory, lookup	Everything fails map-fold transfer (+2 held vs +14–17 train); confidence anti-correlates
D157	Frontier audit	The skipped D109 family-robust objective is the one open branch; D158 frozen
D158	Recurrent q6 PPO relaunch	Stopped by the baseline-dominance audit — the whole D40/q6 world can't beat the resident

14 Arc L — The resident-native pivot (D159-D168, 07-23 → now)

ID	Question	Verdict
D159	Refresh the loss mechanism on 200 games	Catastrophe tail replicated (11% of games, 58% of negative mass); attack angles ranked
D160	Does the resident ever afford worker 3?	Never — zero affordability windows in 195 games; funding is policy, not luck
D161	Same-panel dominance arithmetic	Full q6 oracle only +3.4 (n.s.) vs resident — substrate closed; resident-anchoring becomes law
D162	Bounded reserve/route/commit options	Can't fund worker 3 (5/128 best) — but the crop-safe option envelope is +12.7 with zero regressions
D163	Do fixed components transport?	No — fruit/iron/protection all nonpositive on a disjoint panel
D164	Current-field macro-transition audit	New motif: producer→suppressor→producer cycling, 72% of top-5 games, resident 11%
D165	Return to the remembered crop?	Zero support in 1,024 tasks — the old crop is always gone
D166	Is the return one command?	No: multi-step acquisition journeys, median 16 turns; single-verb controllers closed
D167	Are the journeys regular?	Yes: BANK_SEED frozen-eligible (135/135 local, 71.4% top-5 field); field agents pre-carry seeds through suppression 45%, resident 0%

ID	Question	Verdict
D168	Does executing the return causally help?	No — hand-written successor controllers close (post-return -6.73 , pre-carry -8.21 ; integrity clean); the motif becomes a rollout-valued option for B2.1
D169	Does a unified resident-option envelope clear the +10 gate?	Yes — PASS: $+10.671$ mean, CI $[+9.42, +11.92]$, 65% improved, 0 regressions, tails better than control (100% coverage; every option negative always-on). Opens D170 (Fable-authored closed-loop training design)

15 Inflection points

- **D30** — our generated maps were unlike real maps; weeks of results invalidated, honest substrate built (D33).
- **D40** — a teacher economy good enough to learn from.
- **D97** — joint two-worker coordination has real, measured value.
- **D101** — the gap is production *persistence*, not suppression.
- **D161** — the alternative-economy substrate is weaker than the bot we already have; everything pivots resident-native.
- **D164→D167** — the missing field behavior distilled into one frozen-eligible option.

16 Lesson learnt — what rich players' persistent jobs taught us (D35a and descendants)

D35a decoded two frozen partitions of rich-player replays (a 12-game discovery and a 9-game confirmation split) into **persistent worker jobs**: multi-turn units of intent — “this worker is renewing the farm”, “this worker is felling and banking wood” — rather than per-turn commands. It is one of the program’s foundational results because both the *finding* and the *representation failure* shaped everything after it.

What the decode found. The job vocabulary explains essentially all of what strong players do: direct productive coverage 100% in both partitions, all-unit coverage 98.1%/99.5%, and 96.4%/99.0% of MOVE commands resolvable as travel *for* a job rather than wandering. The median job runs six-to-seven turns — strong play is *committed*, not per-turn reactive. In 62.7%/63.5% of multi-worker turns the workers hold **distinct roles**, and just two job types — RENEW (renewable farm work) and FELL (chop-and-bank) — account for **~94% of all non-idle activity**: the producer/producer/chopper field mechanism in its rawest form.

The representation lesson. The obvious encoding — one flat categorical over whole-team job signatures — failed its own gate: the top 32 team signatures covered only 86.56% of discovery turns against a 90% floor, because teams of one to seven workers generate 176 signatures and a flat head would have to encode *team size* into *class identity*. The verdict distinguished abstraction from representation: **keep the job abstraction, discard the flat joint head, and factorize per worker** — centralized assignment, collision-aware targets, deterministic banking/completion, and a separate global TRAIN decision. That factorized interface became D35b-d (bundles, provenance, repetition), then D97’s validated joint assignments, and ultimately the q6 proposal ABI.

What the lesson did *not* license. Hard-coding the observed roles failed both ways: a forced persistent chopper was inert (D46 — the D40 teacher already chooses FELL_BANK whenever

legal) and a forced persistent producer lost 12 points (D47) — the roles strong players exhibit are *emergent from scheduling*, not rules you can bolt on. And the later field audits sharpened the picture: what actually separates the top cohort is not holding roles but **cycling** them — the same worker produces, suppresses, then produces again (D164, 72% of top-5 games), with production *persistence* through interruptions as the real differentiator (D101: they reap 24% of what they plant; the resident 0.94%). The D-series' current thread — successor jobs, BANK_SEED returns, D168's bounded option — is the direct descendant of D35a's job abstraction applied to that cycling.

17 Model architectures and situation encodings

Three encoding families and half a dozen model families recur across the arcs. This section describes them concretely.

The raw situation. Every encoding starts from the same game state: a rectangular board (up to 22×11 cells) of terrain (soil/water), trees (species PLUM/LEMON/APPLE/ BANANA/IRON-ore analogues, growth stage, hit points, ripe fruit), units (per-troll move speed, carry capacity, hp, chop power, cargo), both players' banked inventories and scores, and the turn number. The referee reveals all of it every turn — the encodings differ in what they make *learnable*, not in what is visible.

Family 1 — spatial planes (the curriculum actor, Arc A). The Rust environment serializes the situation as a **104-channel × 11 × 22 uint8 tensor**. Per-cell planes encode validity (channel 0 doubles as the board mask), terrain, tree species/stage/fruit, and unit positions; scalar facts are *broadcast* as constant planes — e.g. channels 56–61 own inventory and 62–67 opponent inventory (values quantized to uint8 at a $\approx 1/30$ scale), channels 74–78 the selected unit's move/carry/hp/chop/free-capacity stats — and explicit BFS *distance planes* give the network pathfinding for free (added after pure PPO failed to learn it). Actions are likewise spatial: **13 action planes × 11 × 22** — a masked categorical over (action-type, cell), with the legality mask computed by the referee-exact environment.

The curriculum actor architecture (SpatialActorCritic): a 3×3 conv stem 104→16 channels → **four residual blocks** (width 16, two 3×3 convs each) → an actor head (1×1 conv to the 13 action planes, logits flattened and masked) and a critic head (validity-masked global average pool → Linear 16→64 → tanh → Linear→1). Roughly 35k parameters — deliberately tiny, because deployment must fit CodinGame: the accepted pipeline quantizes it to **int8 inside a generated Rust kernel** (max logit drift 0.0000687, 512/512 identical masked choices) and, after the K2/V5 pattern of persistent workspaces and reused buffers, runs a two-worker inference in **7.04 ms median / 17.6 ms warm p95** within a 68,988-byte single-file source. Training used PPO with a 0.10-weight teacher-auxiliary loss (pure PPO deterministically argmax-collapses) and legal-action masking throughout.

Family 2 — scalar situation vectors (selector era). Where a decision attaches to a *moment* rather than a cell, situations are flat feature vectors, always built from observable state only (opponent identity is deliberately excluded — the submitted agent cannot see it): the 56-feature deployable state of D61's option probes, the **64-field state vector** of the q6 program (global economy, workforce, phase, remaining intervention budget, previous-intervention kind), the 139/62/44-feature turn-100 economy models of D63, and D78's spatial-relational add-ons (target condition, worker-to-crop distance, occupancy) that lifted opponent-attack prediction to 0.9307 AUC. The recurring finding: these snapshots *predict behavior* well and *transfer value* poorly — every fitted value model on them collapsed one map-fold away (D153: +14–17 in fit, +1.8 held).

Family 3 — proposal/action vectors (the q6 ABI). Interventions are encoded per *candidate action*: a **379-field proposal vector** (job kind, target class and crop provenance, ETA, predicted deposit and rate, ownership, encoded cell geometry, and the 64-expert endorsement pattern) concatenated with the 64-field state = the 443-feature rows of the D148/D151 corpora. The proposal *generator* is itself a model: the bank of 64 frozen D98 linear scorers, quantized to 6 bits per coefficient (**9,792 coefficients in 9,180 base85 bytes**), whose per-root proposal union retains 86–88% of the exact joint oracle — a compact action basis rather than a policy.

Model families tried on these encodings, smallest to largest: linear probes (224 and 379 weights — D61/D110), a 12-unit echo reservoir with a 52-parameter trained readout (D76), a fully-trained 1,072-parameter recurrent controller (D77), tiny MLPs of 6–7k parameters (D115 ReLU classifier; D153’s state64+action379→16→1 value net), the factorized 6,626-parameter proposal-ranker + 689-parameter act/wait gate (D117–D143), and the 10,725-parameter recurrent shared-proposal policy of D108/D109/D158. The pattern across all of them: capacity was never the binding constraint — representation and objective were. What survived: dense exact-value *teachers* over these encodings (D113, D152), the proposal-union action basis, and the rule that value must be computed at decision time (rollouts) rather than fitted into a snapshot scorer.

18 Why the models kept failing — representation and objective, unpacked

Section 17 ends with a compressed claim: *capacity was never the binding constraint — representation and objective were*. That sentence summarizes roughly sixty failed learning experiments. This section unpacks it.

How we know capacity was not the problem. Models from 52 parameters to 10,725 failed in the *same way*, and adding capacity never moved held-out performance: the 7,121-parameter value net fit its training folds at +14–17 and scored +1.8 one map-fold away (D153); adding feature slices (D154), history memory (D155), or hierarchical lookup (D156) changed nothing. Meanwhile a *parameter-free* hand decoder reproduced all 85,047 of the teacher decisions that a trained MLP captured only 85% of (D41a). The information was present and representable — the models were being asked the wrong question in the wrong language.

Four representation diseases.

- **Wrong function shape for the target.** The teacher’s decision rule is lexicographic — strict priority tiers resolved by exact integer comparisons. Small MLPs express smooth blends of features and approximate hard nested precedence badly at any practical size (D41a: 85% vs the decoder’s 100%). No amount of width fixes a primitive mismatch.
- **Snapshots don’t carry causes.** A feature vector of the current moment encodes *correlates* of value on the training distribution, not the causal game-tree structure that produces value. Hence the sharpest split in the ledger: behavior prediction from snapshots worked beautifully (0.97 AUC for “who will scale”, D63) while value prediction from the same kind of snapshot collapsed under any distribution shift — and model confidence *anti-correlated* with realized value (the top decile predicted +18.07 and realized −1.51; D153b). Same inputs, different question, opposite outcome: a representation boundary, not a data-volume problem.
- **Action spaces too coarse to contain the strategy.** With four semantic modes (balanced/harvest/renew/fell), every optimizer — PPO, CEM, evolutionary lineage search — converged to “always balanced” ($\leq 1\%$ deviations; D73–D77), because the choices that matter (*which* crop, *which* worker, *when exactly*) were not expressible in the vocabulary. The moment actions became concrete jobs (D97), +36.9 of oracle value appeared in the very same games. The state encoding was never the bottleneck there; the *action* encoding was.
- **One scalar forced to mean two things.** The proposal-ranking logit was implicitly asked to encode both relative preference and absolute safety. A translation-invariant ranking loss cannot calibrate absolutes: 82 of 98 losing interventions were ranking errors with a positive arm available at the same decision (D127), and bolting on a separate safety head failed because per-arm false positives compound across many proposals (D129).

Four objective diseases.

- **Pooled margin doesn’t pin down the trade-offs.** Optimizing mean margin over a heterogeneous opponent population lets the learner buy points against one family by selling

another — and *which* trade it makes is undetermined by the loss, so it rotated across independent panels (per-family correlation -0.014 ; D109). Nothing in the objective said “don’t sacrifice anyone.”

- **Margin permits self-harm.** Margin = own – opponent, so suppressing both scores nearly symmetrically looks acceptable to the gradient. Observed literally: -39.98 opponent score bought with -41.74 own (D73, again in D109). The fix — an explicit own-score protection term — was proposed in D109’s closing paragraph and never tested until D170.
- **Hard-argmax targets punish equally good actions.** The teacher picks one of several near-tied moves (70% of states had another action within five points; D152); cross-entropy against the single winner turns near-ties into hard negatives, capping even *training* accuracy near 20% (D149). The objective fought the structure of the task itself.
- **Grading on the wrong distribution.** Supervised imitation grades the policy on the teacher’s states; deployment happens on the policy’s *own* states, which drift immediately — -172.7 paired margin from autoregressive covariate shift (Phases 12–14). Even the *selection* objective had this disease for a while: fit-side regret anti-predicted transfer ($r = +0.89$ in the wrong direction; D131) until selection moved to held-out blocks (D134).

Why D170 is shaped the way it is. Each clause of the current experiment answers one of these diseases: concrete validated options instead of coarse modes (action representation); observable features with recurrent context, and value obtained from actual rollouts rather than fitted snapshots (causal state); a single clean output — invoke or keep (no dual-meaning scalar); paired-control reward to strip map-difficulty variance; closed-loop training so the policy is graded on its own states; and the four-way objective comparison whose whole point is the first two objective diseases — group-DRO against family rotation, an explicit own-score protection term against self-harm — under strictly out-of-fit selection. It is the first experiment in the program where every previously identified failure mode has a specific countermeasure in the design.

19 Why CPU and GPU training never agreed

The YT benchmark that decided where training runs (Arc A, D11 era) found the GPU path **9.8× faster** (9,769 vs 995 transitions/s) — and rejected it anyway: the same frozen hybrid-chopper evaluation scored **52/61 on CUDA versus 56/61 on CPU**, a 6.557-point gap against the preregistered 5-point parity cap, while every other parity metric passed. This is not a bug in either backend; it is the expected physics of the situation:

- **Floating-point addition is not associative.** GPU kernels (cuDNN convolutions, parallel reductions, tensor-core matmuls) sum in different orders than CPU BLAS, and may use different intermediate precisions (TF32, fused multiply-adds), so identical models on identical inputs produce logits differing in the last bits.
- **Reinforcement learning amplifies last-bit noise.** A last-bit logit difference flips an occasional near-tie action; a flipped action changes the trajectory; the changed trajectory changes the training data for every subsequent update. Unlike supervised learning — where the same noise stays bounded — the RL feedback loop compounds it into *macroscopically different policies*. That is exactly the observed signature: bitwise parity on static metrics, divergence only in rolled-out policy outcomes.
- **The same mechanism appears CPU-side with threads.** A 20-thread fit was not byte-stable for one seed (D117), and one threaded model hash differed at $2.4e-5$ drift (D153) — parallel summation order again. Hence the house rules: **one deterministic training thread** for anything selected by byte-identity, thread-parallelism only for *evaluation* whose outputs are compared field-by-field, and byte-identical 1-vs-20 repeats as a standing integrity gate.

The project chose reproducibility over throughput: the sole D11 training run went to local CPU, and the frozen local/YT parity gate remains a precondition before any GPU result may

be *selected* (YT stayed in use for exactly-replayable map/reduce simulation, where byte-parity was achieved and verified per shard).

20 The saved-games database

Everything field-related — loss diagnoses, archaeology, motif audits, opponent models — runs on a locally saved corpus of real arena replays. Its shape as of 2026-07-27:

What we have.

- **8,122 raw replays, 2.4 GB**, in `data/raw/games/` — one JSON per finished arena game, containing the referee’s full record: per-turn frames, both players’ commands and stdout, map layout, and final scores. Every game sits on a distinct map (8,122 unique layouts); **469 distinct agents** appear, including 32 boss games. All 8,122 parse with 0 failures (99.7% exact score reproduction, 0 unexpected mismatches). The 2026-07-28 wide-lens collection quadrupled the corpus in one run (+6,231 games) by fetching the top-20’s FULL visible battle windows and ranks 21-50 for the first time — the earlier 10-per-agent sampling had been leaving ~85% of the rotating stream uncollected.
- **Two immutable D61p snapshots** (`data/raw/snapshots/20260721T105508Z-d61p/`, .../20260727T130712Z-d61p/ — 33 index/manifest files each: leaderboard state at collection time, request and source hashes for every fetch, and the open/sealed partition manifests. Snapshots are append-only; the collector refuses cache overwrite.
- **47 legacy battle records** in `data/raw/battles/` from the Gold-era collector — the previous tooling generation, kept for the historical corpus.
- Derived layers: decoded state streams validated turn-by-turn against the simulator (361,755 transitions, **zero material mismatches**; 28,416 position-only RNG differences), aggregate statistics (`data/processed/stats.json`), and per-experiment extracts (e.g. the D164 field-cycle tables).

How we got them. CodinGame exposes finished games through public replay endpoints: a last-battles list per agent, and one replay JSON per game id. Collection is strictly read-only (GETs only, throttled, hard-stop on HTTP 422/429) and each run is individually authorized, the same discipline as arena writes. Two tool generations: the Gold-era `collect.py` (fetch-log book-keeping into battles/), superseded by the **D61p immutable collector** (`data/scripts/collect_snapshot.py` + `parse_snapshot.py`), which hashes every request and source body, refuses to overwrite cached games, deduplicates into the shared `games/` store (2026-07-27 run: 198 fetched new, 195 skipped as cached, 220/220 requests clean), and physically separates a **sealed confirmation partition** (currently 11 games) that no analysis may open — it is reserved as untouched holdout for future candidate confirmation. Every rebuild passes a QA gate: exact final-score reproduction (1,685 exact + 8 known penalty-only endings), tree invariants, and point-symmetry of every map.

How we group them. The groupings the experiments actually use:

- **By partition discipline** — OPEN vs SEALED-confirmation. The single most important split: sealed games are never decoded by exploratory analyses.
- **By subject** — resident appearances (203 in the current snapshot, 192 open) vs the top-source stratum (exactly 10 appearances per current top-20 agent — a deliberate stratified sample) vs boss games vs the long tail of the 345-agent population.
- **By rank cohort** — top-5 / ranks 6-20 / resident: the standard field split of D164–D167 (e.g. P→S→P cycles: 72% / 27% / 11% of appearances).
- **By outcome** — wins vs losses; within losses, **catastrophes** (margin ≤ -100 ; 19/192 open resident games, carrying 58% of negative-margin mass) vs ordinary losses; early-lead reversals (ahead at turn 100) as their own diagnostic cohort.
- **By matchup** — per-opponent records against the resident (historically the weakest: wala, delineate, norxondor — the 07-16 loss-analysis trio).

- **By behavioral motif** — analysis-derived labels: scaler vs non-scaler appearances (46/104 in D63), renewable-mode games (the yaichi study), worker-rich vs two-worker cohorts, pre-carry vs post-acquisition successor returns (D167).
- **By collection generation and era** — legacy Gold-era battles vs the Legend-era cumulative store vs dated immutable snapshots (which make “the field as of date X” a well-defined, hash-frozen object).

21 Glossary — mother, crop, orchard

Terms that recur throughout the ledger, grounded in the game’s mechanics and the resident’s own source code.

- **Mother** — a tree the bot deliberately plants (or adopts) and then *keeps alive as a renewable seed source* instead of chopping it. Mechanically: a living tree bears species-typed fruit; a picked fruit can be replanted as a seed of that species; chopping the tree instead yields one-time wood. A mother is therefore reproductive capital — its recurring fruit income funds new plantings — where an ordinary tree is harvestable capital. The concept is first-class in the live resident: its scarce-map planner runs the intent chain `NeedSeed → HarvestSeed → PlantMother → TendMother → PlantCrop{mother, target}` (see `rust/src/bin/yamo_orchard_live.rs`).
- **Crop** — a tree planted for *conversion*: grown, then felled for wood/score. In the mother/crop loop, the mother’s fruit becomes the seeds; the crops are her children. A **lineage** is the family of trees descending from one seed source; “lineage extinction” (a denial concept, class d) means no living tree of that family remains.
- **The mother/crop loop** — the self-renewing economy: protect one mature parent, harvest her fruit, plant children, fell the children at maturity, repeat. On tree-sparse maps this loop is the lifeline (the Gold-era deforestation stall was fixed by a seed reserve for exactly this reason). Its cost side: tending a mother consumes worker turns, and a mature *shared* mother’s fruit can be captured by the opponent — the Phase 1-5 verdict that closed six aggressive mother/crop variants was precisely “action cost plus opponent capture exceed private crop value.”
- **Orchard / secure orchard** — the resident’s wrapper that runs this loop on favorable geometry: cells where the mother and her crops sit in resident-controlled territory (the promoted 07-17 stack’s “secure-orchard coverage” widened exactly this geometry). “Releasing” the mother — treating her as a spare resource for other tasks — is closed: the ledger’s verdict is that **the mother is a saturated producer, not an idle reservation** — her fruit throughput is already fully consumed by the loop, so liquidating or re-tasking her trades recurring apples for less wood than they are worth.
- **Where this touches the current thread** — the D164-D168 successor returns are the same economics seen from the worker’s side: a producer leaves to suppress, and comes back by planting a *new crop generation* from a banked seed (D167’s `BANK_SEED`). The deposited bank those seeds come from is fed by harvests — including mother fruit — so the mother/crop loop is the upstream supply of the very returns D168 is now testing.

22 Deep dive — D169, the option-envelope gate (standalone reading)

This chapter explains experiment D169 from scratch, including every experiment it builds on. It assumes no other context. **Update (2026-07-27, post-run): D169 PASSED** — mean envelope +10.671, CI [+9.420, +11.922], 65% of tasks improved, zero regressions, tails better than control, on 100% panel coverage; every option was negative when always-on, so all value is per-game selection. This clears every frozen gate on the first pass (no D169b extension needed) and authorizes designing D170. The rest of this chapter describes the experiment as it was frozen before that result; its reasoning for *why* the gate exists and *why* the thresholds are what they are remains the authoritative explanation.

What D169 is. The live bot (“the resident”) is a hand-written two-worker policy that cannot be beaten by any replacement we have built, yet loses to the top of the ladder in one measured way. The remaining strategy is to keep the resident exactly as it is and add a small vocabulary of bounded, abortable interventions (“options”) on top of it — then *learn when to invoke them*. D169 is the go/no-go measurement for that entire strategy: it computes, on 1,024 replayed games, the **hindsight envelope** — how much better each game could have gone if, with perfect hindsight, we had picked the single best option (or no option) for that game. The envelope is a ceiling, not a policy: if even the ceiling is low, the strategy dies before any training budget is spent on it.

Why this is the remaining strategy (the four results that force it).

- *D159 — the loss mechanism.* On 200 arena games: ~11% of the resident’s games are “catastrophes” carrying ~58% of all lost margin, and in most of them the resident is *ahead at turn 100* before being out-compounded. Replicated independently on fresh data (19/192).
- *D101 — the root cause.* Top agents harvest 24.16% of the crops they plant; the resident harvests 0.94%. Its suppression (chopping enemy crops) is already competitive — what it lacks is *production persistence* while interrupted.
- *D160 — why “just add a third worker” fails.* In 195 decoded games the resident never once accumulates the resources to train a third worker; affordability is a multi-turn funding *policy*, not a windfall you can wait for.
- *D161 — why “switch to a better economy” fails.* The best alternative complete economy we ever built (the D40 teacher family) is 37.8 points *behind* the resident head-to-head, and even a perfect per-game oracle over interventions on that substrate nets only +3.4 (statistically indistinguishable from zero). Verdict: everything must anchor on the exact resident.

What an “option” is. A bounded intervention with exact-resident fallback: it may route *one* worker through a short scripted sequence, holds a horizon (24–32 turns), aborts back to the exact resident on any failure, and leaves every other worker untouched every turn. Options are safe by construction — a game where an option never arms is byte-identical to the resident’s game (verified, not assumed).

The D169 vocabulary, option by option.

- OPT_RETURN — the *successor return*. Lineage: D164 found that in 72% of top-5 games the same worker produces → suppresses → produces again, while the resident does this in 11%. D165 tested “walk back to the crop you remember” — zero support in 1,024 games (the old crop is always gone). D166 showed the real return is a multi-step journey (median 16 turns) starting empty-handed. D167 proved the journey is regular: in 135/135 resident cases and 71.4% of top-5 field cases it is **BANK_SEED** — pick a seed from the deposited bank, walk, plant a new generation. D168 then scripted exactly that as an always-on controller and *lost* (−6.7 to −8.2 paired): forcing the return everywhere is worse than the resident’s own judgment. The surviving question — the one D169 asks — is the per-game one: *are there games where invoking it would have won?* The envelope selects it only where it helps.
- OPT_FRUIT, OPT_IRON, OPT_PROTECT — three *resource-control* options from D162/ D163: temporarily route one worker to fruit harvesting/banking, iron mining/banking, or consumption protection, under a fixed shadow reserve, then return. D163 proved each is *negative on average* when always-on (−2.0 / −3.6 / −0.03) — but D162 measured that the per-task envelope over this family is **+12.7 (CI +9.0 to +16.3) with zero regressions** on a 128-task panel. That asymmetry — bad always, good when selected — is precisely the signature of a timing problem, and timing problems are what the closed-loop program (D170) would learn. D169 re-measures that envelope at full scale (1,024 tasks) with the unified vocabulary.
- TRIG arming — each resource option also gets a variant armed by the *B3.1 trigger*: the audit of catastrophe games showed the resident’s existing endgame switch cannot fire until

the collapse has begun (it requires already being behind), but that the opponent visibly scaling past two workers precedes the collapse by **42-125 turns in 84% of catastrophes**. That observable early warning becomes an arming condition here — making part of the envelope deployability-realistic, not just hindsight.

- *Predeclared extension (D169b)*: if the envelope lands between +5 and +10, one addition is authorized — joint two-worker assignments in the style of D97, which proved (on the old substrate) that coordinating both workers' concrete jobs adds +9.2 beyond the best single-worker action at two-thirds of decision points. It is excluded from D169a only because it must be rebuilt resident-anchored, which costs implementation risk the first measurement doesn't need.

The panel and the integrity discipline. The 1,024 tasks are the consumed D148 evaluation panel: 64 official-generator maps × both seats × eight opponent families, already used by D161/D167/D168 — reusing consumed data is *correct* here because an envelope is an upper-bound measurement, not a selection (nothing is fitted; the prohibition on fitting anything to envelope winners is itself a lesson, from D100b, where hindsight winners proved map- and seat-specific with rank correlations of 0.10-0.22). Every run must: reproduce the resident's 1,024 control games bit-for-bit against D161's frozen records; prove inactive arms byte-identical to control; produce byte-identical results at 1 and 20 threads; and apply crop safety *relative to control* (D122's rule — judging absolute crop counts penalizes games the resident itself fails).

The gates, and why these numbers. Coverage $\geq 60\%$ of tasks armable (else the vocabulary is too narrow to matter). **PASS** needs mean $\geq +10$ with CI lower bound $\geq +5$, $\geq 30\%$ of tasks improved, no negative opponent family, and tails no worse than control. The +10 bar is set against the actual goal arithmetic: the code gap to rank 3 is roughly 4-6 arena points, and every translation step — envelope → learned policy → arena — has historically lost margin (thirty-odd selectors captured under a third of their teachers' value; the one arena promotion came from +4 local). A ceiling below +10 leaves no room for those losses; below +5 (**KILL**) the class cannot even theoretically close the gap, and the honest move is to stop. **BORDERLINE** buys exactly one extension (D169b), never tuning.

What happens after. PASS does not create a bot — it authorizes designing D170: a closed-loop learner over this vocabulary, whose central open question was posed by D108/D109 and never answered: a *family-robust objective with own-score protection* (recurrent policies trained on pooled margin kept "rotating" which opponents they beat, correlation -0.014 across panels, while suppressing their own score). KILL closes Tier 2 of the backlog and the project holds at maintenance. Either verdict is recorded under the same rules as the 168 experiments before it.

Cast of experiments referenced here:

ID	Role in D169
D97	Joint two-worker assignment value (+9.2 beyond best-single) — the D169b extension
D100b	Why nothing may be fitted to envelope winners (hindsight doesn't transfer)
D101	The root-cause diagnosis (reap 24.16% vs 0.94%) motivating the option class
D108/D109	The unanswered objective question D170 would tackle after a PASS
D122	The relative crop-safety measurement rule

ID	Role in D169
D148	Origin of the 1,024-task consumed panel
D159/D160/D161	Its mechanism; funding proof; substrate verdict — why options-on-resident is the only path
D162	The option machinery and the +12.7 envelope precedent
D163	The three resource options; proof they fail always-on (the timing signature)
D164–D167	The successor-return lineage ending in BANK_SEED
D168	Proof that scripting the return always-on loses; per-game selection is the open question
B3.1	The observable opponent-scaling trigger (42–125 turns of warning) used for arming

23 The road ahead — backlog

What we intend to do next, how long each step takes, why it is worth doing, and what we expect from it. The operational source of truth is docs/BACKLOG.md (with gates and kill rules); this chapter is its narrative snapshot as of 2026-07-28.

1. D170b — the closed-loop option policy (in flight; hours of CPU). *Why:* it is the only lever whose measured ceiling (+10.671 hindsight envelope, D169) exceeds the +6.25 gap to the rank-3 bar; every offline alternative was exhausted by thirty-odd failed selectors. *What:* train a $\leq 12k$ -parameter recurrent policy to decide, in-game, when to invoke which bounded option — comparing four training objectives to answer the program’s oldest open question (does family-robustness plus own-score protection prevent the rotation and self-harm that killed pooled-margin training?), then survive out-of-fit selection, a veto panel, and one sealed confirmation run. *Expectation:* honestly uncertain — this is the decisive experiment. Success = $\geq +2.0$ on never-touched maps with no opponent family sacrificed. Both outcomes are valuable: a CONFIRMED opens the deployment path; a CLOSED ends the Tier-2 program with a clean conscience. (The first attempt, D170a, died of a wiring bug caught by its own mechanics gate — the repair is verified and the re-run is executing.)

2. Deployability port (only after a CONFIRMED; ~1 session). *Why:* a confirmed policy is research until it fits CodinGame’s walls. *What:* int8 quantization + persistent-buffer engineering per the proven K2/V5 pattern, $\geq 99\%$ action agreement with the float policy, warm decision $p95 \leq 20$ ms, combined source ≤ 100 KB. *Expectation:* high confidence — the identical pattern already shipped a bigger network at 7 ms / 69 KB.

3. The arena trial (only after everything above; ~1 day including mature reads; requires explicit user authorization). *Why:* the ledger’s hardest lesson — local gates qualify, only arena transfer validates. *What:* the frozen promotion protocol: capacity A/A control, candidate submission, timed reads, keep/revert bands, instant restore on failure. *Expectation:* the honest risk concentrates here — historical local→arena attrition is severe, and the ladder’s slow score recomputation means even a successful promotion shows its true rank over days, not minutes. A confirmed +2–4 local candidate is a *shot* at climbing, not a guarantee.

4. Re-power the field measurements on the quadrupled corpus (1 session, read-only). *Why:* several standing field verdicts rest on painfully small samples — 21 top-5 return cycles,

19 catastrophes, 50 motif appearances — because the old collection lens sampled 10 games per agent. The 2026-07-28 wide collection took the corpus to 8,122 games and 469 agents. *Expectation*: tighter confidence on the motif rates and the catastrophe signature, and better context for judging D170's confirmation numbers before anything approaches the arena. Frozen verdicts stand; this is new evidence, not re-adjudication.

5. Execution-waste sweep on the fresh corpus (1 session). *Why*: execution-class improvements are historically the *only* changes that ever transferred to the arena (+3.0 promotion included). The new corpus contains 29 agents we had never observed. *Expectation*: usually nothing; occasionally a +0.5–1 candidate of the one class with a perfect transfer record.

6. Standing wide-lens collection (minutes to install; awaiting user authorization). *Why*: per-agent battle windows rotate — unharvested games are lost permanently, and the one-shot wide run proved the stream's value (6,231 games in one pass). *What*: a token-free system cron running the throttled, idempotent collector every 12–24 h. *Expectation*: a corpus that compounds by roughly a thousand games per week at zero attention cost.

7. Housekeeping (pull in when convenient): make cargo test --workspace green again by gating one broken research bin (~1 short session); migrate the now-2.4 GB games store to external storage per the established copy-verify-symlink policy (~1 session).

The horizon. If D170b confirms and the arena transfer holds, the next headroom is already mapped: the D169b extension (joint two-worker assignments, +9.2 measured beyond best-single on the old substrate) widens the option vocabulary, and the same closed-loop machinery retrains on it. If D170b closes, the evidence-permitted paths to +6 are exhausted: the project holds at maintenance (standing collection, occasional execution-class sweeps, no arena churn) — and whether to re-scope the goal becomes a decision for the owner, not the ledger.

24 Where the records live

- Ledger vol 1 (Phases + D1–D166, frozen): legend-top3-experiment-cycle-2026-07-18.md.
- Ledger vol 2 (live): legend-top3-experiment-cycle-vol2-2026-07-23.md.
- Per-experiment records: data/analysis/live-agent-6553250/dNNNa--{protocol,lock,result}.
- Closed branches by topic: docs/CONSTRAINTS.md (classes a–h). Live state: docs/STATE.md.
- Priorities and gates: docs/BACKLOG.md.